

Design and Performance Analysis for Indoor Visible Light Positioning With Single LED and Single-Tilted-Rotatable PD

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Abstract—In recent years, single-LED-based visible light positioning (VLP) systems have been developed because of their low complexity, simple configuration, and feasibility for indoor application scenarios. In this article, first, we describe the system model of the single-tilted-rotatable-photodetector (STRP)-based receiver in single-LED scenario considering the first-order reflection signal. Then, we derive the Cramer–Rao lower bound (CRLB) of the positioning error for the received signal strength (RSS)-based positioning system using the discussed model and select proper inclination and rotation angles according to the derived CRLB. In addition, we analyze the least-square (LS) positioning method and propose an improved gray wolf optimization (IGWO)-based localization algorithm with the assistance of the LS method. Finally, the proposed LS-IGWO positioning algorithm has been evaluated in terms of accuracy and latency via software simulations and field tests. Simulation and experiment results show that using the RSS-based positioning model, the proposed LS-IGWO algorithm can achieve centimeter-level positioning accuracy and performs best compared with the original LS, the original GWO, single IGWO, pure LS-GWO, and the existing fingerprint method.

Index Terms—Cramer–Rao lower bound (CRLB), gray wolf optimization (GWO), least square (LS), received signal strength (RSS), single LED, visible light positioning (VLP).

I. INTRODUCTION

NOWADAYS, due to the demand for indoor location-based services, indoor positioning systems (IPSs) have gradually become a research hotspot and attracted plenty of enterprises and institutions to develop commercial products [1]. IPSs aim at real-time and precise tracking of the position of users or objects in many complicated indoor environments, but the global positioning system (GPS) is difficult to achieve that purpose due to its limitations [2]. Therefore, IPSs based on indoor wireless signals, such as Wi-Fi [3], [4], Bluetooth [5], radio frequency identification (RFID) [6], ultrawideband (UWB) [7], and visible light communication (VLC) [8], [9], have been well developed in recent years. Among them, VLC-based IPSs have been paid much attention

because of their high transmission rate, energy conservation, weak electromagnetic interference, long lifespan, and low cost [2].

VLC-based positioning systems usually use one or more LED lamps as the transmitter to help the users get their position. Most existing visible light positioning (VLP) systems use multiple LEDs to achieve high-accuracy positioning because of its strong stability, wide lighting range, and simple receiver structure. However, for a small room, using one LED is sufficient to achieve lighting, so it is a more cost-efficient solution to deploy single LED. On the one hand, for single-LED-based VLP systems, compared with the multi-LEDs, the hardware signal processing delay at the receiver is much lower, because it does not need to use multiple access protocols to distinguish signals from different LEDs, and that improves the real-time positioning performance. In addition, single-LED systems can avoid the design of LED's layout, the interference between LEDs, and the waste of light resources, making it more convenient for the VLP implementation [10]. On the other hand, the designs of high-precision positioning algorithm and appropriate receiver are still challenging for single-LED scenarios. Therefore, in practice, it is necessary to decide to use one or multiple LEDs according to the actual size of the measured room and performance requirements, and VLP systems with single LED can be considered as a cost-efficient IPS solution for a small room.

In VLP systems, received signal strength (RSS) detected by photodetectors (PDs) is more widely used, because they do not need additional synchronization maintenance or high-precision image acquisition devices. Therefore, RSS-PD-based VLP systems can be considered a low-cost solution to the indoor positioning problem [11]. In [12], the impact of the multipath reflection signal on positioning performance was analyzed for RSS-PD-based indoor VLP systems. Moreover, in [13] and [14], the performance limits of LED-based positioning and orientation systems were studied in non-line-of-sight (NLOS) environments. Fingerprint and least-square (LS)-based VLP algorithms considering multipath reflection were studied in [15] and [16], respectively. However, these studies are all implemented in multi-LED scenarios. For single-LED scenes, there are few related studies, so it is necessary to analyze the performance of the VLP system in this case.

At the receiver, PDs or image sensors (ISs) are commonly taken as the device for sensing optical signals [17]. In the

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presence of single-LED-based VLP system, a receiver with multi-PDs, ISSs, or single-rotatable PD is mainly utilized to estimate the position of the target node (TN). In [18], multiple PDs were used in single-LED-based IPS. The receiver is composed of a horizontal PD and two tilted PDs. In [19], the CMOS-made IS was used to estimate the TN's position by finding the camera pose, and they designed a positioning algorithm for an arbitrarily tilted smartphone. Liu et al. [10] designed a receiver with a rotatable PD in single-LED scenario and then proposed a VLP algorithm based on machine learning (ML) methods concerning the first-order reflection of the optical power. In [20], a receiver equipped with a single-tilted-rotatable PD (STRP) was implemented, and accordingly, the fingerprint positioning using K -nearest-neighbor (KNN) method was designed. The experiment was implemented in a $600 \times 600 \times 1135 \text{ mm}^3$ indoor environment, and the average positioning error of 25 TNs is less than 25 mm.

However, some challenges need to be addressed in the existing studies for single-LED scenarios. For example, in [10], the receiver carrying PD is equipped with a long rotation arm, and thus, it is not suitable for system integration, and the fingerprint-based positioning algorithm proposed in [10] has slightly higher complexity. Small-size receivers are designed in [18] and [20], but they did not consider the effect of reflected signals. Therefore, the performance of positioning algorithms presented in [18] and [20] is limited in indoor NLOS environments. Furthermore, there are few studies about single-LED single-PD (SLSP)-based IPS, and the performance of fingerprint-based localization algorithms usually deteriorates when considering the LED orientation. To address the above problems, we further analyze the STRP-based receiver designed in [20] for the SLSP scenario and, accordingly, propose an improved gray wolf optimization (IGWO) positioning algorithm with the assistance of the LS method (LS-IGWO). The performance of the proposed LS-IGWO positioning algorithm is verified via software simulations and field tests. Our major contributions are summarized as follows.

- 1) We derive the Cramer–Rao lower bound (CRLB) of the positioning error using the STRP proposed in [20] considering the first-order reflection signal for single-LED scenario and analyze how to select proper inclination and rotation angles.
- 2) The original LS positioning method based on STRP is derived in single-LED scenario, and then, we design an initial population generation strategy based on the result of the LS method.
- 3) The GWO algorithm is implemented in single-LED scenario to estimate the position of the TN using the fingerprint database generated by the STRP-based VLC channel model.
- 4) We introduce three improvement mechanisms, namely, nonlinear convergence factor, weighted update strategy, and mutation of the head wolf into the proposed LS-IGWO algorithm to address the local optimum problems.
- 5) We evaluate the performances of the original LS [21], the original GWO [22], single IGWO, pure LS-GWO, the fingerprint method [20], and the proposed LS-IGWO

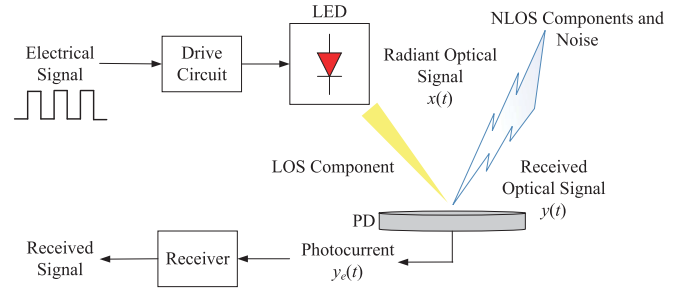


Fig. 1. VLC system diagram.

positioning algorithms in terms of positioning accuracy and latency.

The rest of this article is organized as follows. Section II introduces the VLC channel modeling using Lambert radiation source, STRP-based receiver, and CRLB of the positioning error in the implemented positioning system. Section III derives the original LS method and describes the proposed LS-IGWO positioning algorithm. Section IV analyzes the performance of the proposed LS-IGWO positioning algorithm using software simulation. Section V presents the field test of our proposed method. Finally, our works are concluded in Section VI.

II. SYSTEM MODEL

A. VLC Model

The diagram of the common VLC system is shown in Fig. 1. For the transmitter, electrical signal is converted into optical signal through LED drive circuit, and the amplitude of the optical power signal $x(t)$ denotes the optical power values rather than the electrical-level values. The relationship between $x(t)$ and the received electrical signal $y_e(t)$ is described as follows [23]:

$$y_e(t) = R \cdot x(t) \otimes h(t) + n(t) \quad (1)$$

where R is the PD responsivity and $n(t)$ denotes the additive white Gaussian noise (AWGN) with a double-sided power spectral density (PSD) of $N_0/2$. $h(t)$ is the optical wireless channel impulse response (CIR), including the line-of-sight (LOS) and NLOS components, and can be denoted as follows [23]:

$$h(t) = h_{\text{los}}(t) + h_{\text{nlos}}(t) \quad (2)$$

where $h_{\text{nlos}}(t)$ is composed of multiple reflection CIRs as follows:

$$h_{\text{nlos}}(t) = h_{\text{nlos}^1}(t) + h_{\text{nlos}^2}(t) + \dots + h_{\text{nlos}^l}(t) + \dots \quad (3)$$

where $h_{\text{nlos}^l}(t)$ denotes the l th-order reflection CIR.

The spherical LED-based VLC transmission system is usually described as the Lambert radiation model. In this case, LOS and the first-order reflection components account for a large proportion in the receiver, while the second and higher order NLOS components are negligible [10]. Therefore,

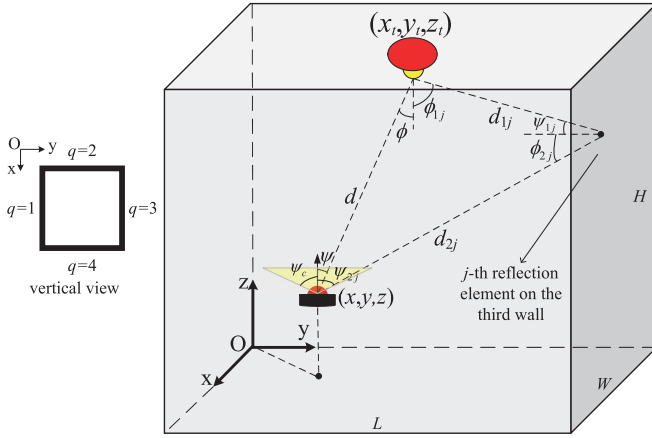


Fig. 2. Single-LED-based VLC channel model.

the simplified radiation model is shown in Fig. 2, and the corresponding CIR can be obtained by [24]

$$h_{\text{los}}(t) = \frac{A(m+1)}{2\pi d^2} G(\phi, \psi) \delta\left(t - \frac{d}{c}\right) \quad (4)$$

$$h_{\text{nlos}^1}(t) = \sum_{q=1}^4 h_{\text{nlos}^1}^{(q)}(t) \quad (5)$$

$$h_{\text{nlos}^1}^{(q)}(t) = \sum_{j=1}^{R_q} \frac{A(m+1)\rho_j A_r}{2\pi d_{1j}^2 d_{2j}^2} G_r(j) \delta\left(t - \frac{d_{1j} + d_{2j}}{c}\right) \quad (6)$$

where A , m , d , ϕ , ψ , and c represent the PD effective area, Lambert order, distance from PD to LED, irradiance angle, incidence angle, and ray velocity, respectively. $\delta(t)$ is the Delta function. A_r is the area of reflection region, and R_q is the number of reflection element on the q th wall. ρ_j , and d_{1j} and d_{2j} are reflection coefficient and distances related to the j th reflection element. $G(\phi, \psi)$ and $G_r(j)$ are defined as follows:

$$G(\phi, \psi) = T_s(\psi) g(\psi) \cos^m(\phi) \cos(\psi) \quad (7)$$

$$G_r(j) = T_s(\psi_{2j}) g(\psi_{2j}) \cos^m(\phi_{1j}) \times \cos(\psi_{1j}) \cos(\phi_{2j}) \cos(\psi_{2j}) \quad (8)$$

where $T_s(\psi)$ and $g(\psi)$ are optical filter gain and PD concentrator gain, respectively. ϕ_{1j} , ψ_{1j} , ϕ_{2j} , and ψ_{2j} are reflection angles related to the j th reflection element. $g(\psi)$ and m can be obtained as follows [24]:

$$g(\psi) = \frac{n^2}{\sin^2(\psi_c)} \quad (9)$$

and

$$m = \frac{-\ln 2}{\ln \cos \phi_{1/2}} \quad (10)$$

respectively. ψ_c , n , and $\phi_{1/2}$ are the PD field of view (FoV) angle, the refractive index of the lens, and the half-power angle of the LED. The formulas related to ψ and ψ_{2j} are valid only when the PD incidence angle is within the FoV angle.

B. RSS Model

At the receiver, we usually process the electrical amplitude signals $y_e(t)$, and the power of $y_e(t)$ can be calculated as follows [23]:

$$P_{re} = \lim_{T_e \rightarrow \infty} \frac{1}{T_e} \int_{-T_e/2}^{T_e/2} y_e^2(t) dt \quad (11)$$

where T_e is the period of $y_e(t)$. According to (4)–(6), the electrical power P_{re} considering the first-order reflection can be modeled as follows [25]:

$$P_{re} = (RP_r)^2 + \sigma_{st}^2 \quad (12)$$

where σ_{st}^2 is the noise power composed of thermal and shot noise, and it can be calculated as follows [25]:

$$\sigma_{st}^2 = \sigma_t^2 + \sigma_s^2 \quad (13)$$

$$\sigma_t^2 = \frac{8\pi k_b T_f}{G_e} \eta_c A_e I_2 B_n^2 + \frac{16\pi^2 k_b T_f \Gamma}{g_m} \eta_c^2 A_e^2 I_3 B_n^3 \quad (14)$$

$$\sigma_s^2 = 2q_e R P_r B_n + 2q_e I_{bg} I_2 B_n \quad (15)$$

where k_b , T_f , G_e , η_c , q_e , I_{bg} , I_2 and I_3 , B_n , g_m , and Γ are Boltzmann's constant, Fahrenheit temperature, open-loop voltage gain, fixed capacitance per unit area of the PD, unit charge, background current, noise bandwidth factors, equivalent noise bandwidth, transconductance of FET, and channel noise factor, respectively. Shot noise includes the influence of ambient light, while thermal noise mainly reflects the influence of electronic devices.

P_t is the total transmitted power of the optical power signal $x(t)$ and P_r represents the received optical power, and they can be obtained as follows [23]:

$$P_t = \int_{-\infty}^{+\infty} x(t) dt \quad (16)$$

$$P_r = P_t [H_{\text{los}}(0) + H_{\text{nlos}^1}(0)] \\ = P_t \left[\frac{A(m+1)}{2\pi d^2} G(\phi, \psi) + \sum_{q=1}^4 \sum_{j=1}^{R_q} \frac{A(m+1)\rho_j A_r}{2\pi d_{1j}^2 d_{2j}^2} G_r(j) \right] \quad (17)$$

Equation (17) is a common RSS model for most VLP systems, and it can be used to establish ranging model [4], [8], fingerprint database [10], [26], or parameter estimation model [27], [28] to obtain the TN's position. Therefore, the signal-to-noise ratio (SNR) can be denoted as follows [23]:

$$\text{SNR} = 10 \lg \frac{P_{re}}{\sigma_{st}^2} \quad (18)$$

When PD and LED are placed horizontally and LED is located at the center of the ceiling (2.5, 2.5, 3), Fig. 3(a) and (b) shows the distributions of the LOS and the first-order NLOS optical power in a $5 \times 5 \times 3$ m³ room. We calculated the received power at 97×97 TNs via simulations, in which some important simulation parameters of the VLP system are shown in Table I. (The subsequent VLP simulations are all based on the parameters listed in Table I.) As a rule of thumb, the first-order reflection components account for about 19.23% of the total optical power and are

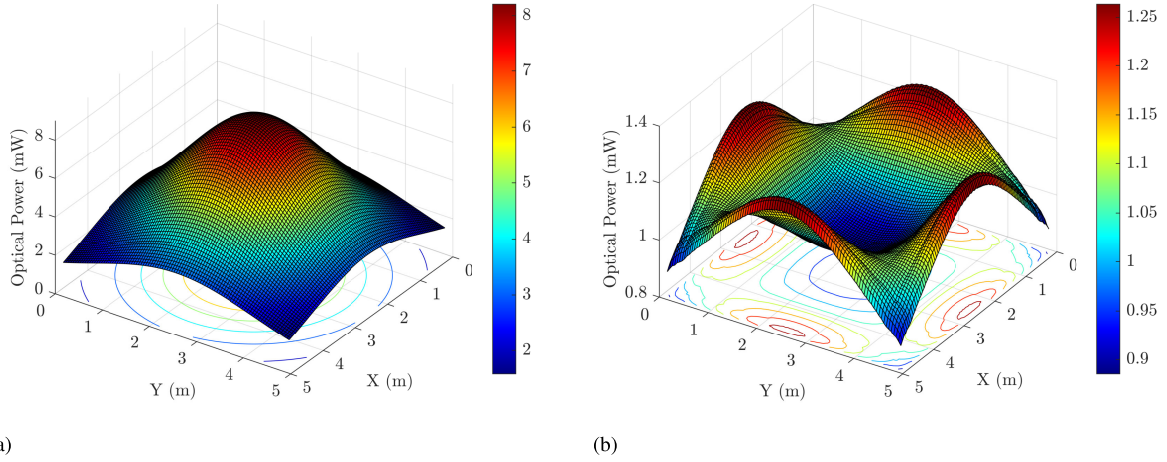


Fig. 3. Distribution of the received power when PD is placed horizontally. (a) LOS path. (b) First-order reflection path.

TABLE I
SIMULATION PARAMETERS OF THE VLP SYSTEM

Symbol	Description	Values
$L \times W \times H$	room size	$5 \times 5 \times 3 \text{ m}^3$
P_t	LED radiation power	100 W
$\phi_{1/2}$	half power angle	60°
ψ_c	PD FOV angle	80°
A	PD effective area	10 cm^2
n	refractive index	1.5
$T_s(\psi)$	optical filter gain	1
(x_t, y_t, z_t)	LED position	(2.5, 2.5, 3) m
z	receiver height	1 m
ρ_j	reflection coefficient	0.8
R_w	number of reflection elements	2250
A_r	area of reflection elements	66.67 cm^2

mainly distributed in the edge area of the simulated room. In this case, the performance of the RSS-based VLP systems may deteriorate if neglecting the reflection signal.

C. Single Rotation Receiver Model

The STRP model used in [20] is shown in Fig. 4. This model uses a receiver that can rotate along its central axis whose direction vector is $[0, 0, 1]^T$ when the receiver's bottom base is placed horizontally, and the PD is tilted on the surface. For this STRP-based receiver in the single-LED scenario shown in Fig. 2, employing the geometric relationships in Fig. 4, the cosine values of ϕ , ψ , ϕ_{1j} , ψ_{1j} , ϕ_{2j} , and ψ_{2j} in (7) and (8) can be obtained as follows:

$$\cos \phi = -\vec{u}_2 \cdot \vec{u}_L \quad (19)$$

$$\cos \psi = \vec{u}_2 \cdot \vec{u}_1^{(i)} \quad (20)$$

$$\cos \phi_{1j} = \vec{u}_{3j} \cdot \vec{u}_L \quad (21)$$

$$\cos \psi_{1j} = -\vec{u}_{3j} \cdot \vec{e}_q \quad (22)$$

$$\cos \phi_{2j} = -\vec{u}_{4j} \cdot \vec{e}_q \quad (23)$$

$$\cos \psi_{2j} = \vec{u}_{4j} \cdot \vec{u}_1^{(i)} \quad (24)$$

where $\vec{e}_q$ is the normal vector of the q th wall and $\vec{u}_L$ is the orientation of the LED, which is $[0, 0, -1]^T$ when the LED is placed horizontally. $\vec{u}_2$, $\vec{u}_{3j}$, and $\vec{u}_{4j}$ are unit direction vectors

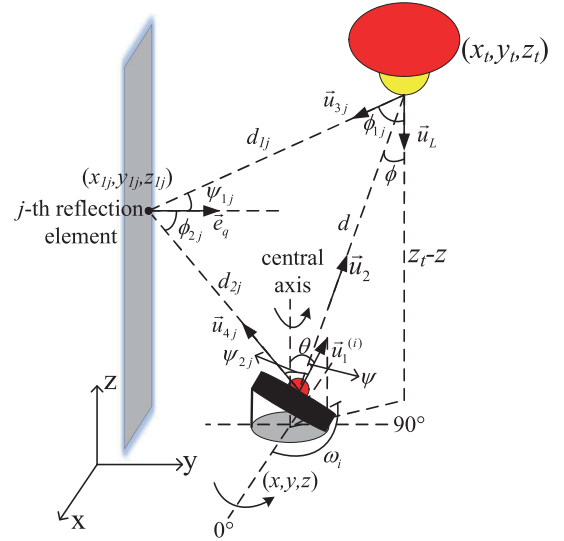


Fig. 4. Illustration of the STRP-based receiver.

related to the LOS and j th reflection element, and they can be obtained by

$$\vec{u}_2 = \frac{(x_t - x, y_t - y, z_t - z)^T}{d} \quad (25)$$

$$\vec{u}_{3j} = \frac{(x_{1j} - x_t, y_{1j} - y_t, z_{1j} - z_t)^T}{d_{1j}} \quad (26)$$

$$\vec{u}_{4j} = \frac{(x_{1j} - x, y_{1j} - y, z_{1j} - z)^T}{d_{2j}} \quad (27)$$

where (x, y, z) , (x_t, y_t, z_t) , and (x_{1j}, y_{1j}, z_{1j}) are the 3-D positions of the TN, LED, and j th reflection element, respectively. $\vec{u}_1^{(i)}$ is the unit orientation vector of PD at i th rotation angle. When the tilt angle is θ and the i th rotation angle is ω_i , according to the Euler rotation matrix, $\vec{u}_1^{(i)}$ can be calculated

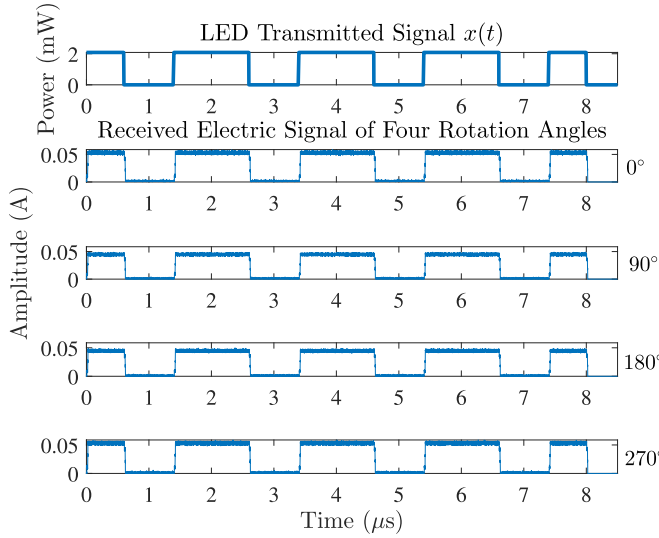


Fig. 5. Simulated power profile of the transmitted optical signal and waveforms of the received electric signal when $N = 4$.

as follows:

$$\vec{u}_1^{(i)} = \begin{bmatrix} \cos \omega_i & -\sin \omega_i & 0 \\ \sin \omega_i & \cos \omega_i & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (28)$$

Assuming that we need to receive N different optical powers at the TN, each rotation angle can be obtained by dividing 360° into N equal parts

$$\omega_i = \frac{(i-1)360^\circ}{N}. \quad (29)$$

Substituting (19)–(24) into (17), the received power of the i th rotation angle can be simplified as follows:

$$P_r^{(i)} = P_t \left[\lambda \frac{\vec{u}_1^{(i)} \cdot \vec{u}_2}{d^{m+2}} + \sum_{q=1}^4 \sum_{j=1}^{R_q} \lambda_{qj} \frac{(\vec{u}_1^{(i)} \cdot \vec{u}_{4j}) (\vec{u}_{4j} \cdot \vec{e}_q)}{d_{2j}^2} \right] \quad (30)$$

where λ and λ_{qj} are independent of the receiver's 2-D position (x, y) , and they can be obtained by

$$\lambda = \frac{A(m+1)}{2\pi} T_s(\psi) g(\psi) (z_t - z)^m \quad (31)$$

$$\lambda_{qj} = \frac{A\rho_j A_r(m+1)}{2\pi d_{1j}^{m+2}} T_s(\psi_{2j}) g(\psi_{2j}) (z_{1j} - z_t)^m \vec{u}_{3j} \cdot \vec{e}_q. \quad (32)$$

Fig. 5 shows the power profile of the transmitted signal and the waveforms of the received signal when using the abovementioned rotatable receiver located at $(2, 3, 1)$ (see Fig. 6). This model is simulated when $\text{SNR} = 30 \text{ dB}$, $\theta = 20^\circ$, and $N = 4$. We use the VLC physical layer modulation method specified in IEEE 802.15.7 [29]. The message code is $[0, 1, 0, 1, 0, 1, 0, 1]$, which is modulated by variable pulse position modulation (VPPM), and the duty cycle of the square wave is 60%. The sampling frequency is 10 GHz, and the width of one symbol is $1 \mu\text{s}$. Because the receiver is located

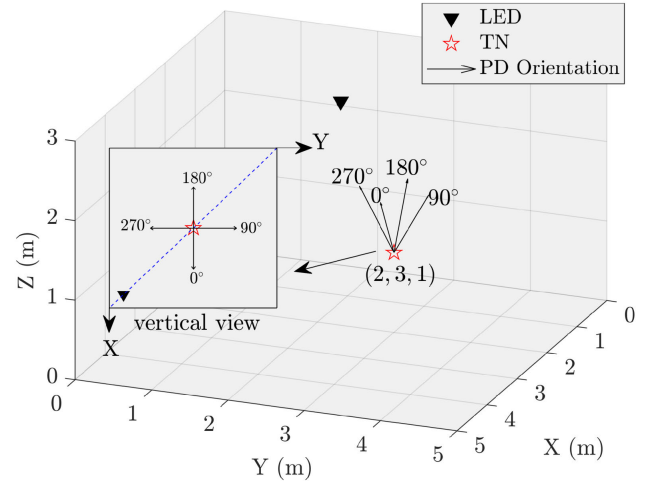


Fig. 6. Diagram of the transmitter-receiver model and PD orientation.

at the diagonal line (diagonal equation: $y = -x + 5$) of the room in 2-D plane, the received signal amplitudes for rotation angles 0° and 270° are approximately equal, and the same applies to 90° and 180° . The PD is facing toward the LED for rotation angles 0° and 270° , and thus, the signal amplitudes are larger compared with the case of 90° and 180° .

For the implementation of an actual VLP system, the STRP model shown in Fig. 4 is placed on a rotatable platform with precise angle scale so as to achieve RSS values detection for different rotation angles. The rotatable platform can be controlled electromechanically, remotely, or manually. In the field experiment, we use a manual rotatable platform to achieve this function. The specific experimental facilities will be introduced in Section V.

D. CRLB of the Localization Error Using STRP

According to the STRP-based VLP system, the actual received power of the i th rotation angle can be expressed as the following random variables:

$$P_{ri} = P_r^{(i)} + X \quad (33)$$

where $X \sim N(0, \sigma^2)$, and the probability density function (pdf) of the received power can be obtained as follows [13]:

$$f_P(P_{ri}) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{[P_{ri} - P_r^{(i)}]^2}{2\sigma^2}}. \quad (34)$$

The received power values of N different rotation angles are independent of each other, and thus, a likelihood function can be obtained by multiplying their PDFs. The likelihood function and its logarithm can be calculated as follows [13]:

$$L(x, y) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{[P_{ri} - P_r^{(i)}]^2}{2\sigma^2}} \quad (35)$$

$$\ln[L(x, y)] = -N \ln(\sqrt{2\pi}\sigma) - \frac{\sum_{i=1}^N [P_{ri} - P_r^{(i)}]^2}{2\sigma^2}. \quad (36)$$

According to the maximum likelihood principle, we can obtain the receiver's position by solving the maximum value

of $L(x, y)$ [13]

$$(x_{\text{est}}, y_{\text{est}}) = \arg \max_{(x, y)} L(x, y) \quad (37)$$

where $(x_{\text{est}}, y_{\text{est}})$ is the estimated position of the TN. A more detailed description about localization method will be introduced in Section III. From (36), the second-order partial derivatives of the logarithmic likelihood function can be obtained as follows:

$$\frac{\partial^2 \ln(L)}{\partial x^2} = \frac{1}{\sigma^2} \sum_{i=1}^N \left(\frac{\partial^2 P_r^{(i)}}{\partial x^2} [P_{ri} - P_r^{(i)}] - \left[\frac{\partial P_r^{(i)}}{\partial x} \right]^2 \right) \quad (38)$$

$$\frac{\partial^2 \ln(L)}{\partial y^2} = \frac{1}{\sigma^2} \sum_{i=1}^N \left(\frac{\partial^2 P_r^{(i)}}{\partial y^2} [P_{ri} - P_r^{(i)}] - \left[\frac{\partial P_r^{(i)}}{\partial y} \right]^2 \right) \quad (39)$$

$$\frac{\partial^2 \ln(L)}{\partial x \partial y} = \frac{1}{\sigma^2} \sum_{i=1}^N \left(\frac{\partial^2 P_r^{(i)}}{\partial x \partial y} [P_{ri} - P_r^{(i)}] - \frac{\partial P_r^{(i)}}{\partial x} \frac{\partial P_r^{(i)}}{\partial y} \right). \quad (40)$$

Fisher information matrix (FIM) is a quantitative representation for the amount of information about online samples, and it can be obtained as follows [13]:

$$I = \begin{bmatrix} -\mathbb{E} \left\{ \frac{\partial^2 \ln(L)}{\partial x^2} \right\} & -\mathbb{E} \left\{ \frac{\partial^2 \ln(L)}{\partial x \partial y} \right\} \\ -\mathbb{E} \left\{ \frac{\partial^2 \ln(L)}{\partial x \partial y} \right\} & -\mathbb{E} \left\{ \frac{\partial^2 \ln(L)}{\partial y^2} \right\} \end{bmatrix} \\ = \frac{1}{\sigma^2} \sum_{i=1}^N \begin{bmatrix} \left[\frac{\partial P_r^{(i)}}{\partial x} \right]^2 & \frac{\partial P_r^{(i)}}{\partial x} \frac{\partial P_r^{(i)}}{\partial y} \\ \frac{\partial P_r^{(i)}}{\partial x} \frac{\partial P_r^{(i)}}{\partial y} & \left[\frac{\partial P_r^{(i)}}{\partial y} \right]^2 \end{bmatrix}. \quad (41)$$

Since localization can be considered as an unbiased estimation problem, the expectation of the squared positioning error can be bounded as follows [13]:

$$\mathbb{E} \{ \| (x, y) - (x_{\text{est}}, y_{\text{est}}) \|^2 \} \geq \text{trace}(I^{-1}) = \text{CRLB}. \quad (42)$$

$\partial P_r^{(i)} / \partial x$ and $\partial P_r^{(i)} / \partial y$ can be calculated by (30). Observing (42), we see that the CRLB decreases with the increase of samples' information.

Fig. 7 shows the curves of the averaged CRLB and PD incidence angle ψ when the PD inclination angle θ varies from 5° to 35° . The number of rotation angles N is 4, and the averaged CRLB is obtained from 576 different positions uniformly distributed in the room. It can be seen that as θ increases, CRLB drops, and at the same time, the positioning accuracy is also improved gradually. This is because when the tilt angle θ increases, the difference between the received power from each rotation angle becomes larger, and that provides more quantity of information about online samples. Therefore, the CRLB gradually declines with the increase of FIM.

However, observing the blue curve in Fig. 7, the incidence angle ψ also increases when θ becomes larger, so it is necessary to set the value of ψ within the PD FoV angle ψ_c . When the receiver is located at (4.8, 4.8, 1) and the rotation angle is 45° (the receiver is at the farthest position and faces away from the LED representing the worst situation in the localization), ψ is within ψ_c as θ varies from 5° to 20° , and that will ensure the PD to precisely detect optical signals in the whole positioning area.

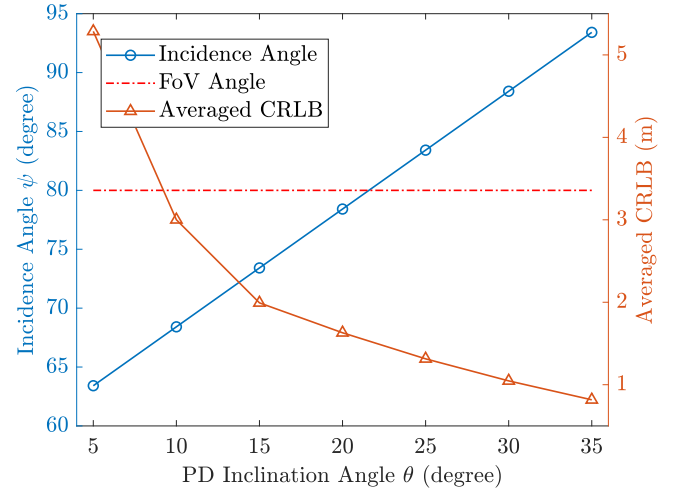


Fig. 7. CRLB of the localization error and PD incidence angle for different θ values.

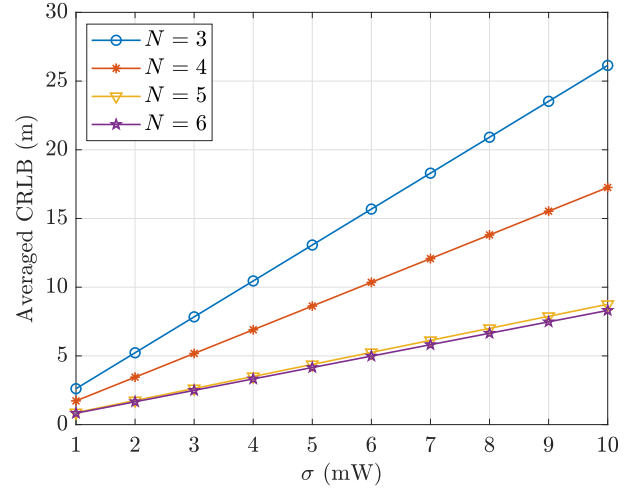


Fig. 8. CRLB of the localization error for different rotation angles with different σ values.

Fig. 8 shows the averaged CRLB with different N and σ values when $\theta = 20^\circ$. We see that as N increases, CRLB gradually drops and becomes very close when $N \geq 5$. This is because a large N value leads to more RSS samples, but the difference between two RSS values becomes smaller, resulting in the saturation of the samples' information. Therefore, five rotation angles may be a good choice. However, we also need to consider the convenience of practical implementation. When $N = 4$, there are four rotation angles, namely, 0° , 90° , 180° , and 270° , and that means we only need to align the rotation reference line with the coordinate axis. Considering both positioning accuracy and practical implementation, we set $\theta = 20^\circ$ for $N = 4$.

III. LOCALIZATION ALGORITHM

A. Original LS Method

Observing (30), the following relationship can be obtained:

$$\frac{P_r^{(i)}}{P_r^{(k)}} \approx \frac{\vec{u}_1^{(i)} \cdot \vec{u}_2}{\vec{u}_1^{(k)} \cdot \vec{u}_2} \quad (43)$$

where $i, k = 1, 2, \dots, N$ and $i \neq k$. Therefore, when N power values are received, M linear formulas can be obtained as follows:

$$\mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{b} \quad (44)$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ \vdots & \vdots \\ a_{M1} & a_{M2} \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_M \end{pmatrix} \quad (45)$$

$$\begin{cases} a_{11} = P_r^{(i)} \sin \theta \cos \omega_k - P_r^{(k)} \sin \theta \cos \omega_i \\ a_{12} = P_r^{(i)} \sin \theta \sin \omega_k - P_r^{(k)} \sin \theta \sin \omega_i \\ b_1 = (P_r^{(i)} \sin \theta \cos \omega_k - P_r^{(k)} \sin \theta \cos \omega_i) x_t \\ \quad + (P_r^{(i)} \sin \theta \sin \omega_k - P_r^{(k)} \sin \theta \sin \omega_i) y_t \\ \quad + (P_r^{(i)} - P_r^{(k)}) (z_t - z) \cos \theta \end{cases} \quad (46)$$

where $M = C_N^2 = N(N-1)/2$. The estimated position of the TN can be obtained by solving the LS solution of (44) [21]

$$\begin{pmatrix} x_{els} \\ y_{els} \end{pmatrix} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}. \quad (48)$$

To obtain the linear equations about (x, y) , the LS-based localization algorithm neglects the reflected optical signal. Therefore, the positioning accuracy of the original LS method is limited.

B. GWO-Based Positioning

To improve the positioning accuracy, we use the swarm-based optimization algorithm. According to (37), solving the maximum of (35) is equivalent to finding the minimum of $\sum_{i=1}^N |P_{ri} - P_r^{(i)}|$. Therefore, we define the objective function as follows:

$$f(\mathbf{w}^{(c)}, \mathbf{P}) = \|\mathbf{P}^{(c)} - \mathbf{P}\|_1 \quad (49)$$

where $\mathbf{P} = [P_{r1}, P_{r2}, \dots, P_{rN}]$ is the N -dimensional actual received optical power vector at the TN and $\mathbf{P}^{(c)} = [P_r^{(1)}, P_r^{(2)}, \dots, P_r^{(N)}]_{(x,y)=\mathbf{w}^{(c)}}$ is the theoretical RSS values of the c th population's position $\mathbf{w}^{(c)}$ obtained by (30). Therefore, the estimated position of the TN can be obtained by solving the minimum value of (49) using the GWO algorithm.

At the t th iteration, the position of the c th wolf individual is updated as follows [22]:

$$\begin{cases} D_\alpha = |2r_2 \times \mathbf{w}_\alpha - \mathbf{w}^{(c)}(t)| \\ D_\beta = |2r_2 \times \mathbf{w}_\beta - \mathbf{w}^{(c)}(t)| \\ D_\delta = |2r_2 \times \mathbf{w}_\delta - \mathbf{w}^{(c)}(t)| \end{cases} \quad (50)$$

$$\begin{cases} \mathbf{w}_1 = \mathbf{w}_\alpha - (2ar_1 - a) \times D_\alpha \\ \mathbf{w}_2 = \mathbf{w}_\beta - (2ar_1 - a) \times D_\beta \\ \mathbf{w}_3 = \mathbf{w}_\delta - (2ar_1 - a) \times D_\delta \end{cases} \quad (51)$$

$$a = 2 - \frac{2t}{T} \quad (52)$$

$$\mathbf{w}^{(c)}(t+1) = \frac{\mathbf{w}_1 + \mathbf{w}_2 + \mathbf{w}_3}{3} \quad (53)$$

where $\mathbf{w}_\alpha$, $\mathbf{w}_\beta$, and $\mathbf{w}_\delta$ are the positions of the wolves with the best, the second best, and the third best fitness at each iteration. r_1 and r_2 are two independent random variables uniformly

TABLE II
GENERATED ARTIFICIAL DATABASE

$N \times S$ Optical Power Values					Position
$P_{lr}^{(1)}$	$\dots$	$P_{lr}^{(i)}$	$\dots$	$P_{lr}^{(N)}$	(x_1, y_1)
$\vdots$	$\dots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
$P_{sr}^{(1)}$	$\dots$	$P_{sr}^{(i)}$	$\dots$	$P_{sr}^{(N)}$	(x_s, y_s)
$\vdots$	$\dots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
$P_{Sr}^{(1)}$	$\dots$	$P_{Sr}^{(i)}$	$\dots$	$P_{Sr}^{(N)}$	(x_S, y_S)

distributed in the interval $[0, 1]$. T is the total number of iterations. At the end of the iteration, the estimated position of the TN can be obtained by

$$(x_{\text{est}}, y_{\text{est}}) = \arg \min_{\mathbf{w}} f(\mathbf{w}, \mathbf{P}). \quad (54)$$

C. Proposed LS-IGWO Positioning Method

For the GWO algorithm used in the abovementioned VLP system, three problems need to be addressed: fitness calculation, initial population generation, and local optimum. The solutions to these problems are stated as follows.

1) *Artificial RSS Database*: For (49), the calculation of $\mathbf{P}^{(c)}$ is inconvenient, because a large amount of reflected optical power in (30) will increase the complexity of the GWO iteration. Therefore, in the offline phase, we use (30) to generate an artificial database shown in Table II. When we need to calculate an individual's fitness, we can use its position to obtain the theoretical values of the N received powers prestored in the database.

In fact, when the RSS parameter estimation model is simple enough, there is no need to establish a database, such as Table II. We only use the database to accelerate the simulation of the GWO algorithm instead of proposing a fingerprint-based positioning method. In our field test shown in Section V, we will use a newly measured RSS model for localization in small areas.

2) *Initial Population Generation Strategy Based on LS*: To improve the convergence accuracy, the estimated position of the TN by LS method is used to generate the initial population in this article. As shown in Fig. 9, the four vertices of the generated area (V_1, V_4) , (V_2, V_4) , (V_1, V_3) , and (V_2, V_3) can be obtained as follows:

$$\hat{\omega} = \arg \min_{\omega} [P_{r1}, P_{r2}, P_{r3}, P_{r4}] \quad (55)$$

$$\begin{cases} (V_1, V_2, V_3, V_4) = (x_{els} - D, L, y_{els} - D, W), \hat{\omega} = 0^\circ, 90^\circ \\ (V_1, V_2, V_3, V_4) = (0, x_{els} + D, y_{els} - D, W), \hat{\omega} = 90^\circ, 180^\circ \\ (V_1, V_2, V_3, V_4) = (0, x_{els} + D, 0, y_{els} + D), \hat{\omega} = 180^\circ, 270^\circ \\ (V_1, V_2, V_3, V_4) = (x_{els} - D, L, 0, y_{els} + D), \hat{\omega} = 270^\circ, 0^\circ \end{cases} \quad (56)$$

where D can be determined by the positioning error of the LS algorithm. By using (55) and (56), an area enclosing the TN can be generated. Therefore, generating the initial population of the GWO algorithm in this region can effectively narrow the search of the population and improve the positioning accuracy.

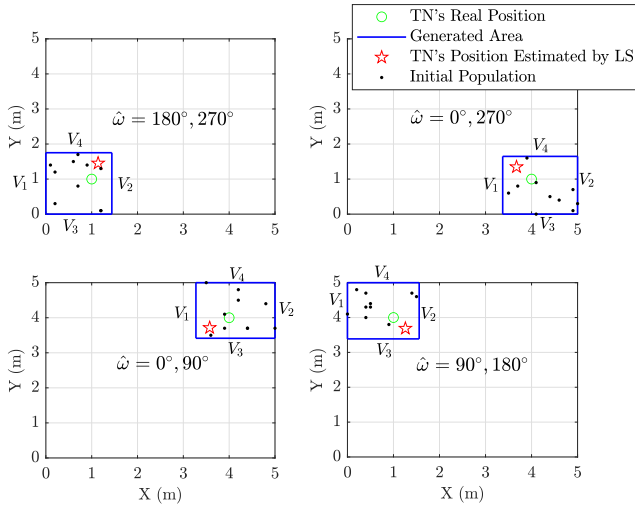


Fig. 9. Illustration of the population generating area.

3) *Improved GWO*: We design three methods to avoid the local optimum of the original GWO algorithm: nonlinear convergence factor, weighted updating strategy, and mutation of the head wolf. We use the following equations to optimize the balance parameter a and the updating process of wolves

$$a = \frac{4}{1 + (t/T)^2} - 2 \quad (57)$$

$$\mathbf{w}^{(c)}(t+1) = \frac{F_\alpha \times \mathbf{w}_1 + F_\beta \times \mathbf{w}_2 + F_\delta \times \mathbf{w}_3}{F_\alpha + F_\beta + F_\delta} \quad (58)$$

where F_α , F_β , and F_δ are the fitness values of $\mathbf{w}_\alpha$, $\mathbf{w}_\beta$, and $\mathbf{w}_\delta$. Equation (57) ensures that at the beginning of the iteration, the wolves search for prey with a larger step while with a smaller step at the end of the iteration. In (58), a weighted average instead of the original arithmetic average is used to capture prey more accurately. Furthermore, to avoid the α wolf falling into the local optimum, we adopt the following operations to adjust the position of the α wolf

$$\mathbf{w}'_\alpha = \mathbf{w}_\alpha + r \times \|\mathbf{w}_\alpha - \mathbf{w}_\beta\|_1 \quad (59)$$

$$\text{if } f(\mathbf{w}'_\alpha, \mathbf{P}) < f(\mathbf{w}_\alpha, \mathbf{P})$$

$$\mathbf{w}_\alpha = \mathbf{w}'_\alpha$$

$$\text{else } \mathbf{w}_\alpha = \mathbf{w}_\alpha \quad (60)$$

where r is a random variable uniformly distributed in the interval $[-1, 1]$.

IV. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed LS-IGWO positioning algorithm in terms of accuracy and latency when LED is placed horizontally and tilted on the ceiling, respectively. During the simulations, the **averaged positioning error (APE)** is obtained from 576 different positions uniformly distributed in the simulated room illustrated in Fig. 2, and for all the positioning algorithms related to GWO, the number of wolves and iterations is set to 25 and 10, respectively. The VLC signals shown in Fig. 5 are used in the simulations, and all the parameters about VLP are listed in Table I.

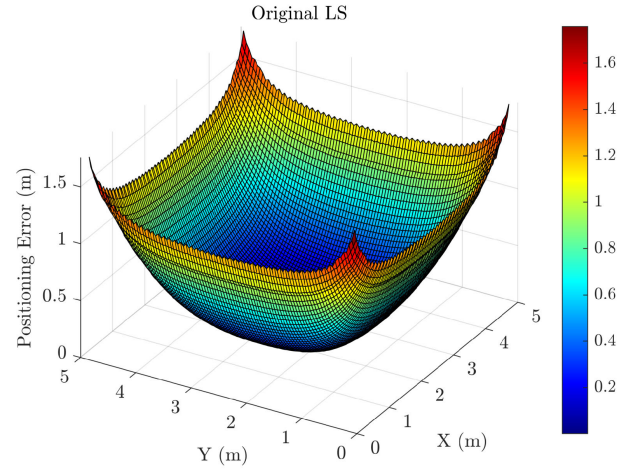


Fig. 10. Positioning error distribution of the original LS method using the STRP-based receiver.

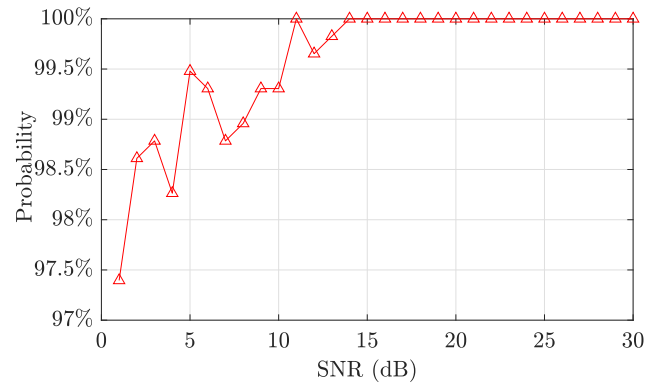


Fig. 11. Probability of enclosing the TN using the generated area for different SNR values.

A. Case in Which LED Is Vertically Downward

Fig. 10 shows the positioning performance of the original LS [21] positioning algorithm for each investigated TN in the whole room when $\text{SNR} = 20$ dB. It can be seen that due to the influence of reflected optical power near the edge area, the positioning accuracy becomes worse. The average positioning error in the edge and center area of the room is 0.80 and 0.16 m, respectively, and the maximum positioning error is 1.76 m (we consider that the area in which the horizontal and vertical coordinates range from 1 to 4 m is the central area, while the rest is the edge area). Therefore, neglecting the NLOS components is unreasonable, and thus, the performance of the original LS method is unsatisfactory when it is used directly.

Fig. 11 shows the probability that the area generated by (55) and (56) can enclose the TN at each investigated position for different SNR values when $D = 2$ m. We see that due to the dependence on the performance of the LS method and the noise of the received optical power, the generated area may not contain TN for low SNR values. However, when $\text{SNR} > 14$ dB, the region can enclose TN with a probability of nearly 100%, so the improvement for the original GWO in terms of

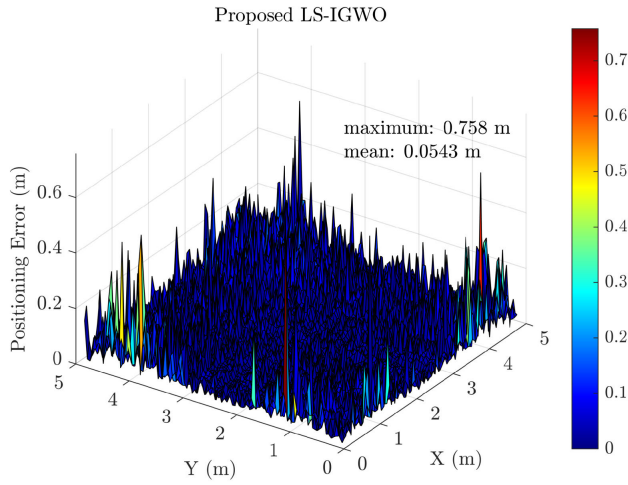


Fig. 12. Positioning error distribution of the proposed LS-IGWO positioning algorithm.

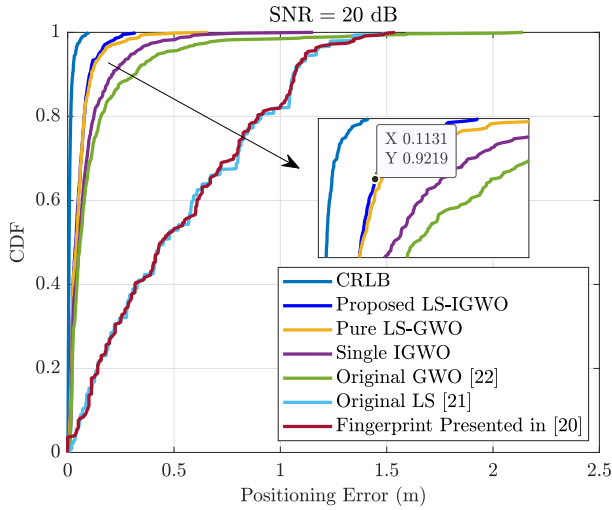


Fig. 13. CDF of the positioning error for the six investigated positioning algorithms.

positioning accuracy would make more sense for higher SNR values.

Fig. 12 shows the overall positioning performance of the proposed LS-IGWO localization algorithm under the same conditions as Fig. 10. We see that compared with Fig. 10, the overall positioning error has significantly decreased by using the proposed LS-IGWO algorithm. According to our observation, the positioning error in the edge and center areas of the room has decreased by 92.13% and 73.42%, respectively, and the maximum positioning error drops to 0.758 m.

Fig. 13 shows the positioning error's cumulative distribution function (cdf) of the original LS [21], original GWO [22], GWO with the initialization strategy shown in (55) and (56) (pure LS-GWO), GWO using the improvement mechanism given by (57)–(60) (single IGWO), the proposed LS-IGWO, and the fingerprint method in [20] for SNR = 20 dB. We see that the positioning error of the proposed LS-IGWO algorithm is closest to the theoretical CRLB boundary, and about 92.19% of the positioning error values are within 0.1131 m for the

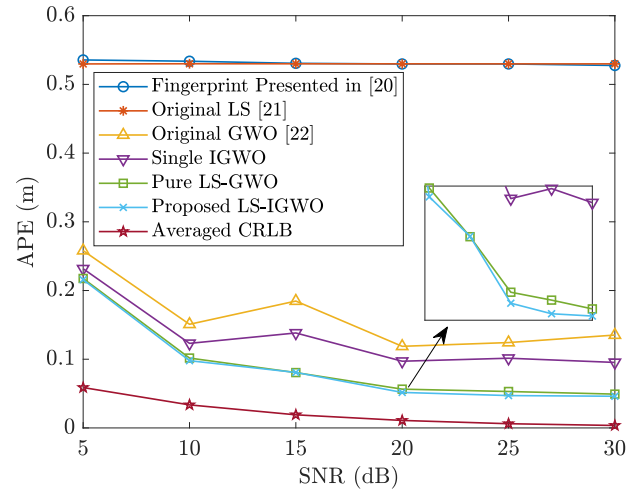


Fig. 14. APE of the six investigated positioning algorithms for different SNR values.

proposed LS-IGWO algorithm. Compared with the original GWO algorithm, the positioning accuracy has improved by about 59.36% using the proposed LS-IGWO algorithm.

Fig. 14 shows the comparison of the six investigated algorithms in terms of the APE under different SNR values. We see that the proposed LS-IGWO positioning algorithm has better positioning accuracy than other algorithms at all the investigated SNR values. Because the original LS and the fingerprint localization algorithms in [20] do not consider the influence of the reflected optical signal, their positioning performance is unsatisfactory. In addition, we can also see that the single IGWO and pure LS-GWO perform better than the original GWO [22]. Therefore, the initial population generating strategy (55) and (56) and the improvement mechanisms (57)–(60) are effective for the proposed LS-IGWO algorithm.

Fig. 15 shows the time consumption of the six investigated algorithms. We see that compared with the original LS [21] and GWO [22] algorithms, the proposed LS-IGWO algorithm can enhance positioning accuracy at a limited cost of time complexity. For the proposed LS-IGWO algorithm, the combination of the original LS and GWO methods in conjunction with the improved strategies increases the time complexity. Therefore, the proposed LS-IGWO algorithm can be considered for the scenarios that are insensitive to time delay. Because the original LS [21], original GWO [22], single IGWO, pure LS-GWO, and the proposed LS-IGWO algorithms do not require online fingerprint database matching, the positioning latency is much lower than the fingerprint method in [20] using KNN.

Fig. 16 provides the APE and positioning latency of the proposed LS-IGWO algorithm with different numbers of wolves when SNR = 20 dB. We see that when the number of populations is more than 25, the improvement in the positioning error becomes smaller and smaller, so we use 25 populations in our simulations. As the number of populations increases, the positioning accuracy becomes better, but the positioning delay also becomes larger. Therefore, we can adjust the population size according to the actual positioning requirements

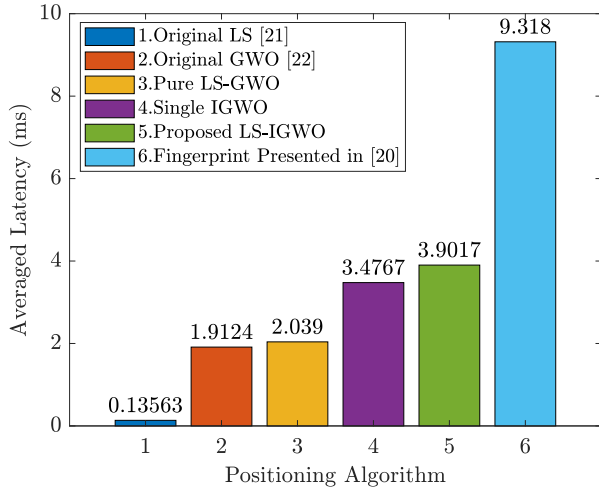


Fig. 15. Time consumption of the six investigated positioning algorithms.

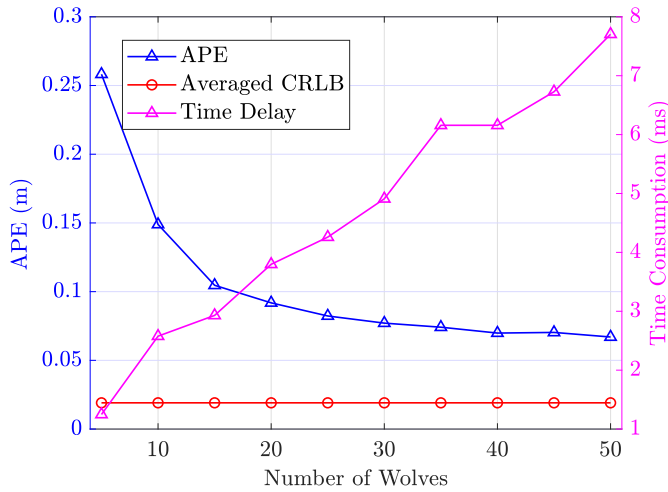


Fig. 16. APE and time consumption of the proposed LS-IGWO positioning algorithms with different population sizes.

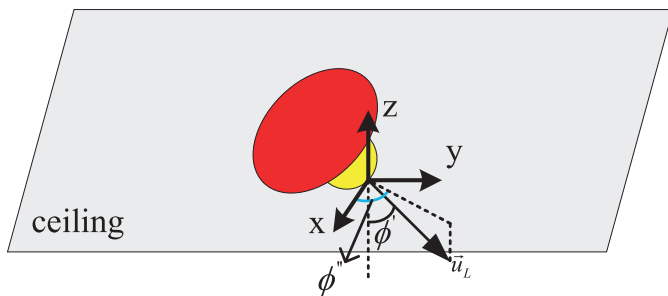


Fig. 17. Illustration of the LED orientation.

to achieve a trade-off between the computational complexity and the positioning accuracy.

B. Case Considering the LED Orientation

In practical scenarios, the orientation of LEDs may change because of environmental changes, and that can affect positioning performance. Therefore, we need to study the influence of LED orientation on positioning accuracy. LED orientation usually refers to the maximum radiation direction of the

optical power. In the previous section, we assumed that the orientation of the LED is perpendicular to the ceiling, i.e., $\vec{u}_L = [0, 0, -1]$. As shown in Fig. 17, the orientation of the LED can be represented by elevation angle ϕ' and azimuth angle ϕ'' , and thus, $\vec{u}_L$ can be calculated as follows:

$$\vec{u}_L = \begin{bmatrix} \cos \phi'' & -\sin \phi'' & 0 \\ \sin \phi'' & \cos \phi'' & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \phi' & 0 & -\sin \phi' \\ 0 & 1 & 0 \\ \sin \phi' & 0 & \cos \phi' \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}. \quad (61)$$

Therefore, by updating the cosine values of the irradiance angles ϕ and ϕ_{1j} according to (19) and (21), we can obtain the following STRP-based RSS model:

$$P_r^{(i)} = P_t \left[\lambda' \frac{(-\vec{u}_2 \cdot \vec{u}_L)^m (\vec{u}_1^{(i)} \cdot \vec{u}_2)}{d^2} + \sum_{q=1}^4 \sum_{j=1}^{R_q} \lambda'_{qj} \frac{(\vec{u}_1^{(i)} \cdot \vec{u}_{4j}) (\vec{u}_{4j} \cdot \vec{e}_q)}{d_{2j}^2} \right] \quad (62)$$

where λ' and λ'_{qj} are independent of the receiver's position (x, y, z) and can be obtained by

$$\lambda' = \frac{A(m+1)}{2\pi} T_s(\psi) g(\psi) \quad (63)$$

and

$$\lambda'_{qj} = \frac{A \rho_j A_r (m+1)}{2\pi d_{1j}^2} T_s(\psi_{1j}) g(\psi_{1j}) (\vec{u}_{3j} \cdot \vec{u}_L)^m (\vec{u}_{3j} \cdot \vec{e}_q). \quad (64)$$

Therefore, after updating the RSS model $P_r^{(i)}$ according to (62), the TN's position can be estimated in the presence of different values of LED orientation by employing the positioning algorithm introduced by Section III.

Fig. 18 shows the distribution of the received optical power in the whole room when $\phi' = 15^\circ$ and $\phi'' = 90^\circ$. From Fig. 18, we see that compared with the results presented in Fig. 3, the peak of the LOS optical power pattern is shifted toward the orientation of the LED, and because the maximum radiation direction of the LED is toward the third wall ($q = 3$), the first-order NLOS optical power received near this wall increases. In other words, most of the received power will be distributed near the main beam of the LED.

Fig. 19 shows the overall positioning error distribution of the proposed LS-IGWO positioning algorithm for different ϕ'' values when $\phi' = 15^\circ$ and $\text{SNR} = 20$ dB. We see that under different LED orientations, the maximum and mean values of the positioning error vary from 0.75 to 0.90 m and from 0.05 to 0.07 m, respectively, and they are very close to that shown in Fig. 12. Therefore, based on the updated RSS model in (62), the proposed LS-IGWO positioning algorithm has good robustness to the LED orientation.

Fig. 20 shows the cdf of the positioning error about the six investigated positioning algorithms when $\phi' = 15^\circ$, $\phi'' = 90^\circ$, and $\text{SNR} = 20$ dB. It can be seen that the proposed LS-IGWO algorithm also has the highest positioning accuracy, and the curve of the cdf is close to the result in Fig. 13. Because the fingerprint positioning presented in [20] did not consider

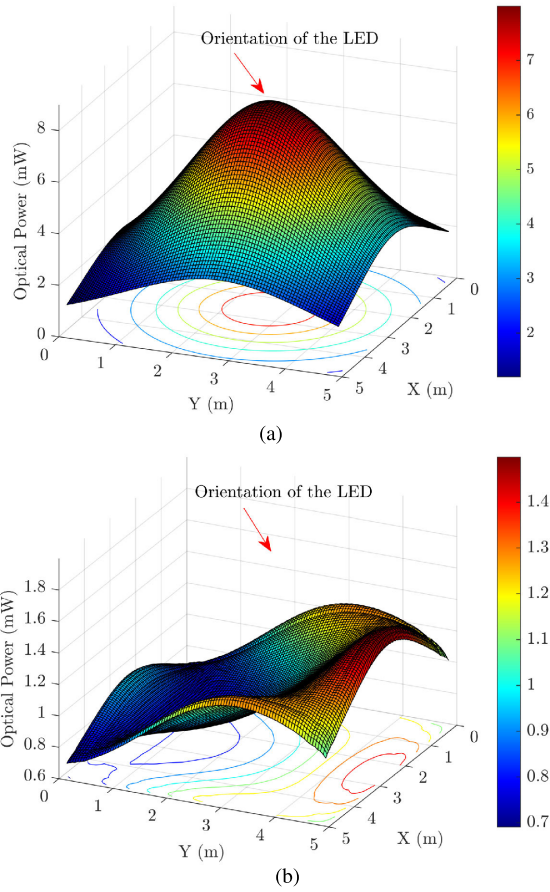


Fig. 18. Distribution of the received power when PD is placed horizontally for $\phi' = 15^\circ$ and $\phi'' = 90^\circ$. (a) LOS path. (b) First-order reflection path.

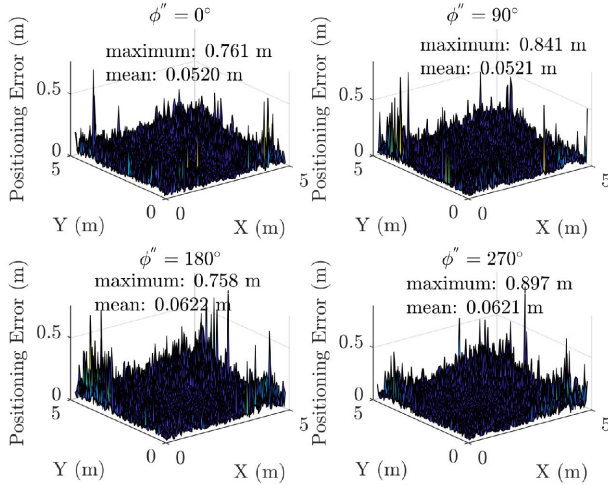


Fig. 19. Positioning error distribution of the proposed LS-IGWO positioning algorithm when $\phi' = 15^\circ$ and SNR = 20 dB for different ϕ'' values.

the LED orientation, its positioning accuracy becomes worse compared with Fig. 13.

In conclusion, constructing an RSS-based parameter estimation model with a known LED orientation can achieve almost the same positioning accuracy as the case that the LED is vertically downward, as long as the PD can detect the optical powers at each rotation angle. Therefore, it is necessary to

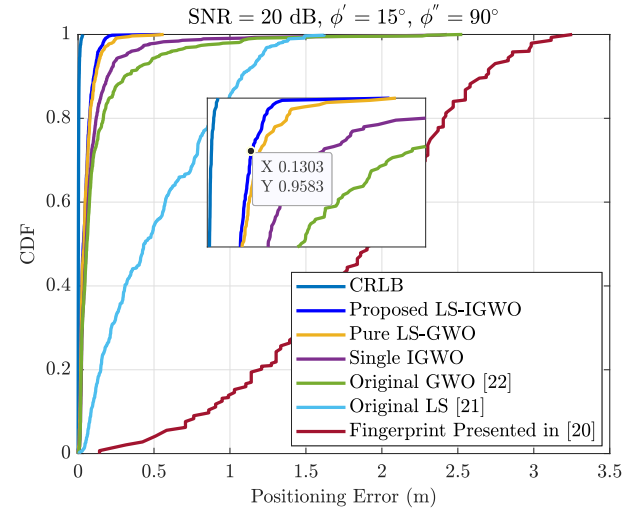


Fig. 20. CDF of the positioning error for the six investigated positioning algorithms (considering the LED orientation).

optimize the layout of indoor facilities to ensure that at each TN, there is always an LOS link between LED and PD as much as possible. When the LED orientation is unknown, in order to achieve sufficiently high positioning accuracy, beamforming techniques can be used in VLC system to detect the maximum radiation direction of the LED, and that is also a challenging issue for our future.

V. FIELD EXPERIMENTS

In this section, we present some field tests to verify the effectiveness of the proposed LS-IGWO positioning algorithm. As shown in Fig. 21, the previous simulated room is proportionally scaled down to a measured area of $50 \times 50 \times 30$ cm³. A single-LED lamp (3 W with a chip produced by Bridgelux) is installed on the center of the ceiling, and the baseband signal generated by Arduino is modulated to the optical wave through the LED drive circuit. In actual measurement, we placed one side of the experiment platform against the wall, thus considering the reflection optical power from this wall.

At the receiver, the PD (Model S6968) is placed on the center of the slope of the slanted cut cylinder (the slanted angle is 20° , and the bottom diameter is 2 cm), and they are fixed on a rotatable platform. We use a manually operated rotation platform REG25 with a scale value of 2° , and the height of the assembled receiver is 1.864 cm. In practical applications, we can use electromechanically controlled rotation platforms, such as model Thorlabs CRMIP, to obtain the rotation angle ω_i more precisely.

As shown in Fig. 22, for practical applications, we can fix the assembled receiver on the center of the tracked object. Assuming that the angle between the directions of the object's principal axis and the x -axis is ω' , in the initial state, the orientation of PD is consistent with the direction of the object's principal axis, i.e., $\omega_i = \omega'$. When the positioning process begins, the receiver first rotates ω' clockwise detecting the RSS values from 0° and then rotates 90° , 180° , and 270° counterclockwise to obtain the RSS values of the rest three

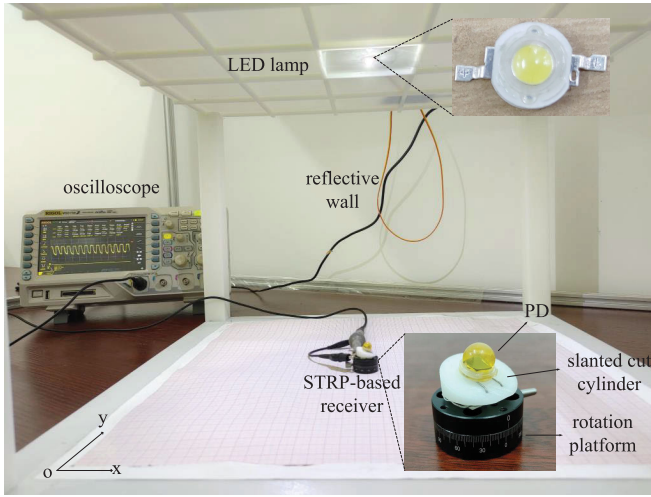


Fig. 21. Layout of the indoor SLSP-based VLP environment.

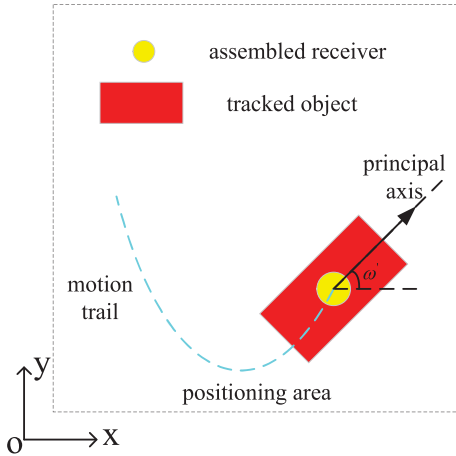


Fig. 22. Diagram of the STRP-based object tracking for practical applications.

rotation angles. Therefore, the orientation of the tracked object relative to the measured room needs to be known in advance, and that is a limitation of the STRP-based positioning system in terms of practical implementation.

In this practical scenario, only the distance d and rotation angle ω_i are related to the received power. Based on our field measurements and using curve fitting method, the following RSS model can be obtained:

$$P_r^{(i)} = C_2 \frac{\vec{u}_1^{(i)} \cdot \vec{u}_2}{d^{C_1}} \quad (65)$$

where $C_1 = 1$ and $C_2 = 1.8012$. Because (65) is obtained from an actual environment, C_1 and C_2 are parameters related to light source, optical devices, reflection optical power, ambient light, and noise. For different distance and rotation angle values, the measured RSS samples, fitted RSS model (65), and Lambert radiation model are shown in Fig. 23. We tested 12 RSS values for different rotation angles 0° – 360° at a given position (15, 15) cm and 15 different positions with a given rotation angle 0° . It can be seen that the RSS value nonlinearly decreases with the distance and varies approximately

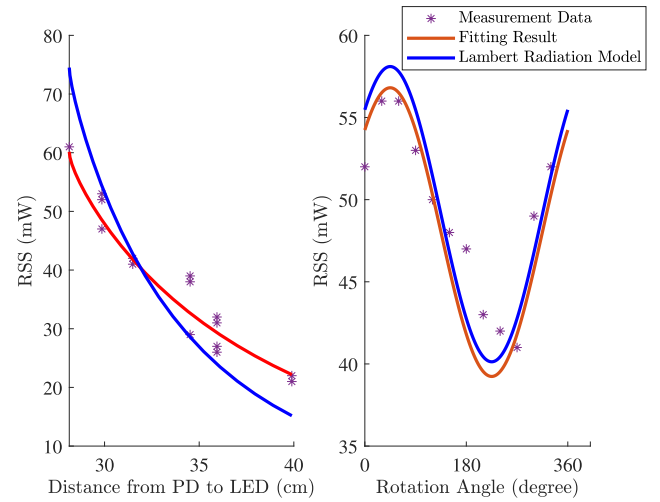


Fig. 23. Sampled RSS values, fitted curves, and Lambert radiation model for different distances and rotation angle values.

TABLE III
RMSE BETWEEN RSS SAMPLES AND LAMBERT MODEL AS WELL AS FITTED CURVE

	Lambert Model	Fitting Result
Distance-RSS	22.93 mW	3.68 mW
Rotation Angle-RSS	7.96 mW	8.44 mW

according to the cosine value of the rotation angle, and that is approximately consistent with the general Lambert radiation model.

The root-mean-square error (RMSE) between the measured RSS samples and the above two model curves in Fig. 23 are shown in Table III. We see that the fitted model (65) is generally more accurate than the Lambert radiation model used in the previous simulation. Actually, in practical positioning scenarios, not all LED lamps can be modeled as a standard Lambert radiation source because of the various shapes of lamps and indoor facilities. We usually need to obtain the RSS model based on actual measured RSS samples and mathematical tools, such as curve fitting. Therefore, in the practical localization, we recommend the newly derived RSS model (65) instead of the original simulation model. When considering the complex NLOS environments, we can update C_1 and C_2 in (65) according to the RSS samples newly measured in that case.

Therefore, by using (65), we can construct the parameter estimation model, such as (37), and the TN's position can be estimated by the positioning algorithm demonstrated in Section III. For good robustness, it would be better to regularly update the RSS model based on changes in indoor environments, such as furniture layout and ambient light.

Fig. 24 shows the overall positioning effect of the proposed LS-IGWO algorithm using the RSS data measured at 25 positions uniformly distributed in the experimental environment. The value of D is adjusted to 3 cm, and other parameters of the positioning algorithm are consistent with the previous simulation. It can be seen that the proposed LS-IGWO algorithm has good positioning accuracy and robustness. Due

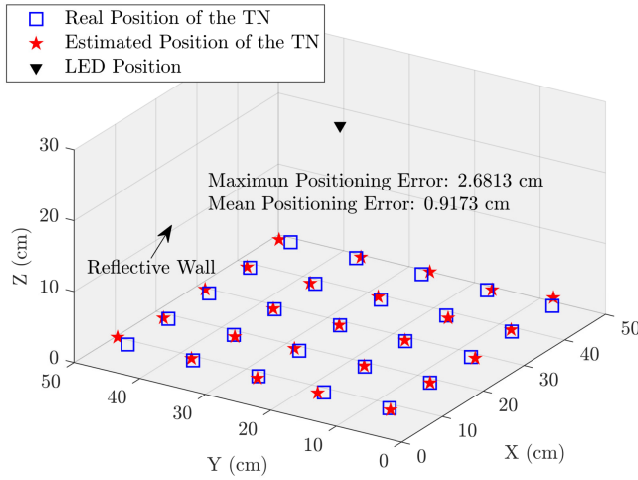


Fig. 24. Positioning result of the proposed LS-IGWO algorithm (field test result).

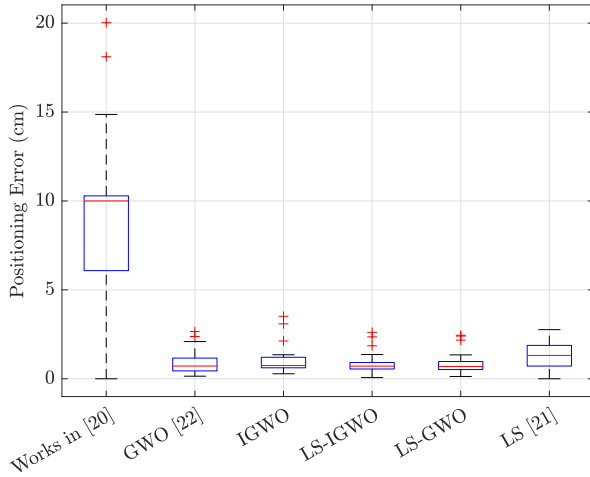


Fig. 25. Boxplot distribution of the positioning error for the six investigated positioning algorithms (field test result).

to the influence of the reflection optical power, the APE of a column against the wall is 1.65 cm, which is slightly higher than other positions.

Fig. 25 shows the minimum, first quartile, median, third quartile, and maximum positioning error values of the six investigated algorithms. These results further verify the effectiveness and robustness of the proposed LS-IGWO algorithm for indoor positioning.

In addition, according to our statistics, the averaged calculation time of all the GWO-based localization algorithms is about 9.7 ms. Due to the simplicity of the RSS parameter estimation model established by (65), there is no need to establish a fingerprint database, such as previous simulations, to improve computational efficiency.

VI. CONCLUSION

In this article, we evaluated the theoretical performance of an existing STRP-based receiver in the SLSP scenarios and then proposed an LS-IGWO-based VLP algorithm to mitigate the impact of the first-order reflection optical signals. According to the derived CRLB of the positioning error, the proper

angles of inclination and rotation are selected. Simulation and field test results show that for different LED orientations, the proposed LS-IGWO positioning algorithm has higher positioning accuracy and stronger stability than the original LS [21], the original GWO [22], single IGWO, pure LS-GWO, and the existing fingerprint positioning algorithm [20]. Therefore, the proposed LS-IGWO positioning algorithm with STRP is effective and feasible and can be considered as a competitive solution for SLSP-based IPS.

In our future work, we will focus on developing RSS-based VLP systems, including more accurate single-LED-based positioning schemes, real-time positioning algorithms, simultaneous positioning and orientation for the receiver, LED orientation estimation, and system implementation for 6G applications.

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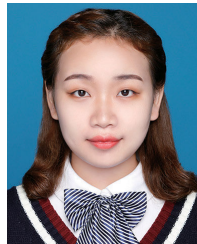
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